Paleomagnetism and structural history of the Ghost Range intrusive complex, central Abitibi Belt, Ontario: further evidence for the Late Archean geomagnetic field of North America

JOHN WM. GEISSMAN,[1] DAVID W. STRANGWAY, AND ANN M. TASILLO-HIRT[2]
Department of Geology, University of Toronto, Toronto, Ont., Canada M5S 1A1
AND
LARRY S. JENSEN
Precambrian Geology Division, Ontario Geological Survey, 77 Grenville St., Toronto, Ont., Canada M5S 1B3

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On the basis of their remanence properties, the lithologies of the Late Archean (2710–2703 Ma) Ghost Range Complex, an east–west-trending layered mafic–ultramafic extrusive sequence in the central Abitibi Greenstone Belt, can be divided into three groups. Group 1 units contain a high-coercivity, high-blocking temperature (greater than 520°C) magnetite-dominated remanence characteristic of the complex (D = 280°, I = 2°, k = 5.5, α_{95} = 11.8°, virtual geomagnetic pole = 13°E, 7°S; isolated by both AF and thermal methods), in good agreement with the few previous results bearing on the Late Archean apparent polar wander path for North America. Group 2 units contain a low-coercivity, low-blocking-temperature (less than 310°C) scattered remanence residing in pyrrhotite. Often, both remanences coexist in a single lithology at a given site. Group 3 units contain distributed coercivity and blocking-temperature remanences, again residing in magnetite, that are more scattered but statistically identical to the mean group 1 direction. The order of magnetic blocking appears to have been group 1 before group 3 before group 2. The geologic setting of the Ghost Range suggests that it has remained essentially stable since emplacement and therefore the group 1 direction appears to reliably represent a Late Archean paleomagnetic pole.

En utilisant les propriétés de rémanence, les lithologies du complex Ghost Range d'âge archéen supérieur (2710–2703 Ma), une séquence extrusive de roches mafiques–ultramafiques, de direction est–ouest, de la ceinture centrale de roches vertes de l'Abitibi, peuvent être subdivisées en trois groupes. Les unités du groupe 1 sont caractérisées par une force coercitive élevée, une température de blocage élevée (dépassant 520°C), et une rémanence portée principalement par la magnétite caractérisant le complexe (D = 280°, I = 2°, k = 5,5°, α_{95} = 11,8°, pôle géomagnétique virtuel = 13°O, 7°S; isolée par deux méthodes, CA et thermique), s'accordant avec les résultats antérieurs peu nombreux au sujet de la dérive apparente des pôles en Amérique de Nord à l'Archéen supérieur. Les unités du groupe 2 sont caractérisées par une force coercitive faible, une basse température de blocage (moins de 310°C), et une rémanence dispersée et portée par la pyrrhotite. Souvent, ces deux rémanences coexistent au sein d'une même lithologie à un endroit donné. Les unités du groupe 3 présentent une force coercitive et une température de blocage variables, les rémanences étant portées à nouveau par la magnétite, elles sont plus dispersées mais correspondent statistiquement à la direction moyenne du groupe 1. L'ordre dans le blocage magnétique semble avoir passé par le groupe 1 avant le groupe 3 avant le groupe 2. Le contexte géologique du complexe Ghost Range suggère qu'il est demeuré stable depuis sa mise en place et par conséquent la direction de rémanence du groupe 1 semble représenter avec fiabilité un pôle paléomagnétique pour l'Archéen supérieur.

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Introduction

In other discussions (Geissman *et al.*, in preparation; Tasillo-Hirt *et al.* 1982), the paleomagnetism of basaltic to rhyolitic volcanics (Kinojevis and Blake River Groups) of the central, least altered portion of the Abitibi Greenstone Belt is shown to be complex. These volcanics appear to have acquired their natural remanent magnetization (NRM) over a significantly long period of time following emplacement and consequently were generally considered to be unreliable recorders of the Late Archean field. In some instances, however, components of magnetization in the volcanics do appear to record a Late Archean field. The coarse-grained equivalent of these volcanics, including sills, stocks, and "flows" of ultramafic to mafic composition, is one of the few rock types available in the central Abitibi Belt for study of the Late Archean field. Other studies have concentrated on the latest Archean – earliest Proterozoic north–south-trending Matachewan dikes (Strangway 1964; Fahrig *et al.* 1965; Irving and Naldrett 1977; Ernst and Halls 1980; Schutts and Dunlop 1981). Study of the

[1]Present address: Department of Geology, Colorado School of Mines, Golden, CO 80401, U.S.A.

[2]Present address: Institut für Geophysik, ETH-Hongger-burg Ch-8093, Zurich, Switzerland.

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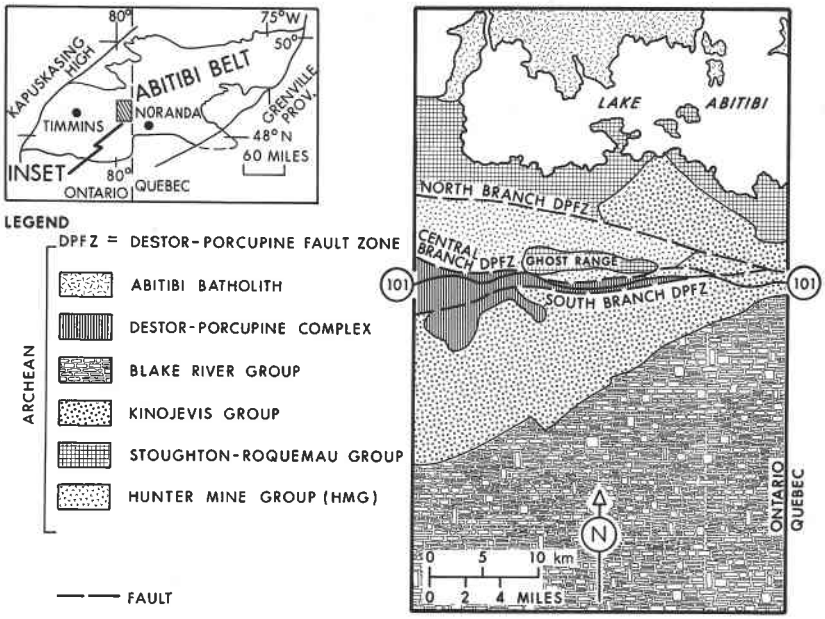


FIG. 1. Geologic map and location map of the Ghost Range Complex area, modified after Jensen (1981).

remanence in these basic intrusives and extrusives may provide data on the Late Archean field, as well as on the magnetic mineralogy and subsequent alteration. In this paper, we discuss magnetization data from one such mafic–ultramafic complex in the central Abitibi Belt, the Ghost Range.

General geology

The area sampled (Fig. 1) is part of Goodwin and Ridler's (1970) Noranda–Benoit volcanic complex, in the central portion of the Abitibi Greenstone Belt, a latest Archean supracrustal terrane in the Superior Province. The Ontario portion of the complex is composed of three subalkalic volcanic cycles, each associated with volcanogenic sedimentation and intrusion of comagmatic units (Jensen 1978, 1981).

The Ghost Range Complex is one of a series of east–west-trending layered ultramafic complexes situated just north of the main branch of the Destor–Porcupine Fault Zone (Fig. 1). These complexes are genetically related to basal komatiites and basaltic komatiites of Jensen's (1979) Stoughton–Roquemaure Group. The Stoughton–Roquemaure Group constitutes the lower third of the second subalkalic volcanic cycle in the central Abitibi Belt (Jensen 1978, 1979; Geissman *et al.*, in preparation). Tholeiitic volcanics of the Kinojevis Group and calc-alkaline volcanics and hypabyssal intrusives of the Blake River Group, whose paleomagnetism is discussed by Geissman *et al.* (in preparation), Tasillo-Hirt *et al.* (1982), and Schutts (1980), overlie volcanics and sediments of the Stoughton–Roquemaure Group. The Ghost Range itself unconformably overlies calc-alkaline volcanics (Hunter Mine Group) of an earlier volcanic cycle. Finally, the complex stands out as an isolated magnetic anomaly; values are in excess of 4000 γ (4 μT) (Letros *et al.*, in preparation) and consequently the complex has received some attention from an exploration standpoint.

The age of the Ghost Range Complex is relatively well bracketed. Rhyodacitic units of the Hunter Mine Group in Rand Township west of the complex yield a U–Pb zircon age of 2710 ± 2 Ma, and units at the top of the Blake River Group yield a value of 2703 ± 2 Ma (Nunes and Jensen 1980). These values are slightly younger, however, than the 2765 ± 47 Ma value reported by Zindler *et al.* (1978) on komatiites and related intrusives of the Stoughton–Roquemaure Group in Munro Township dated by the samarium–neodynium method. Matachewan dikes, which cut all of the Archean rocks in the area, have been dated by Rb–Sr methods at 2690 ± 93 Ma (Gates and Hurley 1973).

The petrology and geochemistry of ultramafic–mafic complexes associated with classic komatiitic flows of Munro Township about 16 km to the west of the Ghost Range have been well described (Arndt 1977; Pyke *et al.* 1973). By comparison, few descriptions of the Ghost Range have been published. Satterly (1951, 1953) mapped part of the complex and suggested that it was in a west-plunging syncline. He believed that it was injected prior to folding of the surrounding volcanics and subsequently underwent differentiation into a gabbro–diabase upper portion and a peridotitic lower portion.

Recent geologic mapping of the Ghost Range (Fig. 2*a*) and a more complete understanding of the regional stratigraphy and structural history have allowed a reinterpretation of the Ghost Range structure (Jensen 1981). Cross sections of the complex (Fig. 2 *b* and *c*) indicate that it is a nearly horizontally layered series (north dip of less than 5°) of coarse-grained extrusive units. These units were emplaced on what was to become in latest Archean time a structural high resting between two major branches of the then active Destor–Porcupine Fault Zone, as a result of subsidence of volcanic basins on the north and south sides of the two faults. Subsidence in the basins occurred with the accumulation of additional lavas belonging to the Stoughton–Roquemaure Group as well as younger Kinojevis and Blake River Groups. The Ghost Range Complex remained a structually high feature sandwiched between major branches of the Destor–Porcupine Fault and was neither significantly tilted nor rotated, whereas the volcanics in the adjoining basins became folded into east-trending synclinoria (Jensen 1981). Other intrusive complexes, farther west in and near Munro Township, were tilted into steeply dipping positions (Arndt 1977) by movement along en échelon faults of the Destor–Porcupine Fault Zone and subsidence. A prehnite–pumpellyite grade of burial metamorphism for the Ghost Range units implies that the complex could never have been overlain by the 40 km or more of younger volcanic and sedimentary stratigraphy in this part of the Abitibi Belt. Because of the implied simple structural history of the Ghost Range, we have focussed our attention on this particular complex.

Mapping by Satterly (1951, 1953), MacRae (1969), and Jensen (1981) has delineated the Ghost Range rock units. Their distribution is shown in Fig. 2 and in ascending order the units are: Hunter Mine Group (HMG, Fig. 2), folded calc-alkalic andesite and rhyolite; peridotite (1, Fig. 2), of variable thickness often in excess of 100 m and containing irregular zones of dunite; pyroxenite (2, Fig. 2), 2–6 m thick, composed mainly of clinopyroxene; gabbro (3, Fig. 2), about 60 m thick, magnesium-rich, enstatite-bearing; cherty tuffs (4, Fig. 2) 0.1–1.0 m thick; and gabbro (5, Fig. 2), up to 50 m thick, iron-rich, augite-bearing. Above the cherty tuffs, the iron-rich gabbros consist of relatively rare fine-grained tabular bodies of black gabbro (sheeted dikes and sills, Fig. 2*c*) alternating with coarse-grained bodies of dark green gabbro.

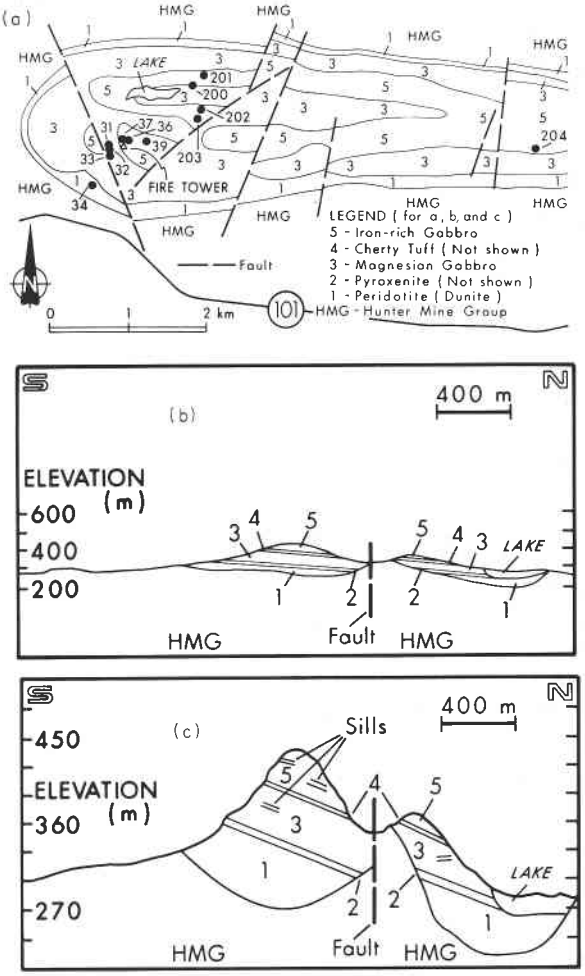


FIG. 2. (*a*) Generalized geologic map of the Ghost Range Complex showing locations of sampling sites (circles). (*b*) Cross section of the complex, no exaggeration. (*c*) Cross section, approximately 7× vertical exaggeration to show relationships among units. Modified after Jensen (1981).

Paleomagnetic sampling and methodology

All four rock types (ultramafics: serpentinites, Mg-gabbros, Fe-gabbros, and fine-grained gabbro sheets) in the Ghost Range were sampled, with the greatest emphasis on the iron gabbros because they are best exposed. A total of 93 samples were collected from 12 sites, shown in Fig. 2*a*. Inaccessibility, overburden, and a dense bush prevented more complete sampling. Samples were collected by hand (areas of very poor access) or with portable drilling equipment. Topographically high and very well exposed areas were purposely avoided, as anomalously high NRM intensities, suggestive of lightning-struck material, were encountered in preliminary work. Generally, two to six specimens (2.5 cm diameter × 2.5 cm height) were prepared from each sample.

Measurements of the magnetization were carried out, usually, with a cryogenic magnetometer (4 in. (10.2 cm) diameter vertical access), but a portable Schonstedt magnetometer (model PSM-1) and a Schonstedt spinner magnetometer (model SSM-1a) were used to measure remanences of moments greater than 0.25 A m^{-1} (2.5 ×

TABLE 1. Representative rock magnetic and remanence properties, Ghost Range Complex rock types

Sample	Rock type	H_C (mT)	H_{CR} (mT)	J_s (A m^{-1})*	J_r/J_s*	T_C (°C)	X_i	NRM (A m^{-1})	$H_{\frac{1}{2}}$ (mT)	T_B (°C)
37A	Fe-gabbro, pyrrhotite-free (G-1)	6.8	78.0	—	0.10	575	6.0×10^{-5}	0.0047	26.0	510–580
37G	Fe-gabbro, pyrrhotite-free (G-1)	8.0	90.0	110	0.036	—	8.6×10^{-5}	0.016	45.0	300–585
37J	Fe-gabbro, pyrrhotite-free (G-3)	—	—	—	—	580	3.0×10^{-4}	0.61	7.0	150–560
36C	Fe-gabbro, pyrrhotite-free (G-1)	—	—	—	—	573	6.2×10^{-5}	0.0032	59.0	100–530
34A	Serpentinite (G-3)	6.1	30.0	920	0.069	574	4.2×10^{-3}	0.25	7.0	50–540
200E	Mg-gabbro, pyrrotite-bearing (G-2)	—	—	—	—	305, 564	6.1×10^{-5}	0.14	18.0	100–310
201E	Fe-gabbro, pyrrhotite-bearing (G-2)	—	—	—	—	280, 560	4.7×10^{-5}	0.0093	3.0	25–570
201F	Fe-gabbro, pyrrhotite-bearing (G-2)	15.0	27.0	—	0.068	310, 550	4.4×10^{-4}	0.190	—	—
202G	Fe-gabbro, pyrrhotite-free (G-1)	8.0	—	—	—	—	3.0×10^{-5}	0.0029	44.0	200–570
202J	Fe-gabbro, pyrrhotite-bearing (G-2)	—	—	—	—	265	9.0×10^{-5}	0.500	—	130–310
202L	Fe-gabbro, pyrrhotite-bearing (G-2)	2.2	—	—	—	300	8.1×10^{-5}	0.360	3.0	80–310
203B	Fe-gabbro, pyrrhotite-free (G-3)	6.8	27.0	—	0.16	575	4.8×10^{-5}	0.031	9.0	200–560
204F	Fe-gabbro, pyrrhotite-bearing (G-2)	9.3	17.0	—	0.28	280, 500	5.2×10^{-5}	0.019	14.0	80–310

NOTES: H_C = coercivity; H_{CR} = coercivity of remanence; J_s = saturation magnetization; J_r/J_s = ratio of saturation remanence to saturation magnetization; T_C = Curie temperature; X_i = initial susceptibility; NRM = intensity of the natural remanent magnetization; $H_{\frac{1}{2}}$ = median destructive field; T_B = blocking temperature range of the remance decaying to the origin in thermal demagnetization.

*J_s and J_r/J_s values were often difficult to determine accurately because of the high (5×10^{-5} or greater) paramagnetic susceptibility of many samples.

10^{-4} emu cm^3). Specimens were demagnetized in alternating fields (AF) with a single-axis Schonstedt AF demagnetizer (model GSD-1). Thermal demagnetization was carried out with either a Schonstedt thermal demagnetizer (model TSD-1) or a non-inductively wound, home-built cylindrical furnace around a glass tube capable of holding five specimens in a vacuum in a field maintained at less than 10^{-4} Torr (0.013 Pa).

All components of the NRM in each sample were resolved using vector subtraction on vectors indicated by progressive demagnetization diagrams (Zijderveld 1967; Dunlop 1979). Each vector was calculated by a linear least-squares fit. In some cases, at high peak fields in AF demagnetization, additional (spurious) magnetization components were acquired and, in order to accurately resolve the true remanence of the sample, the detailed AF demagnetization procedure of French (1976) was employed. In this procedure, it is assumed that an anhysteretic remanent magnetization (ARM) is acquired parallel to the last axis demagnetized. By demagnetizing the sample six times at the same field along both the positive and negative directions of each axis, the ARM can be subtracted.

Results

The paleomagnetic data from the Ghost Range are best discussed in terms of the behavior of individual samples or groups of samples upon demagnetization treatment, rather than by site, geographic location, or rock type. Representative magnetization data from the Ghost Range are given in Table 1. As can be seen, behavior of the magnetization often varies substantially on the site level, within a given lithology. Histograms of NRM intensities and susceptibilities of all rock types are given in Fig. 3. Rock magnetic properties are summarized in Table 1.

Group 1 samples

Samples in this group are generally Fe-rich gabbros and contain a magnetization component of west to northwest declination and shallow positive or negative inclination (Fig. 4). A few components are essentially antipodal to these. In general, NRM intensities for these samples are relatively low, ranging between 0.0029 and 0.035 A m^{-1}. The westward component invariably resides in high-coercivity, high-blocking-temperature (540–580°C) phases as shown in Fig. 5 *a–c*. In each case, a shallow-inclination, westward component is isolated and decays smoothly to the origin after the removal of secondary components. Both thermal and AF demagnetization usually serve to isolate this component.

The isolation of a west-declination component is best exemplified by samples from sites 36, 37, and 203 (Fig.

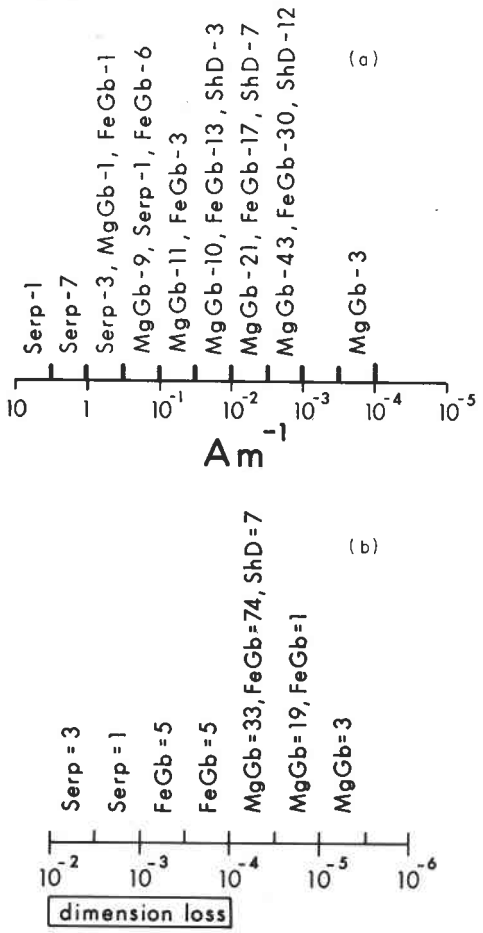


FIG. 3. (*a*) Histogram of NRM intensities, Ghost Range rock types. (*b*) Histogram of initial susceptibilities, Ghost Range rock types. Note that the number of specimens of each rock type measured for NRM and susceptibility data is different; only the general distribution of values is of concern.

6) but several samples from other sites, as will be discussed shortly, do possess this component and show identical behavior in demagnetization. Because the west-declination component is of low dispersion and is found in all rock units, it is concluded that this is a magnetization characteristic of the Ghost Range.

Secondary components of magnetization present within group 1 samples are strongly dispersed (Fig. 4*b*) and consequently must correspond to younger thermal overprinting over a span of geologic time. For example, samples from sites 36 and 37, five scattered outcrops of Fe-gabbro north of the Ghost Range fire tower (Fig. 2*a*), yielded some of the data (identified) shown in Fig. 4 *a–c*. The westward-declination component is clearly characteristic of the majority of the samples, but secondary components are present.

Group 2 samples

Samples from this group, both Fe- and Mg-rich gabbros, are characterized by significantly higher NRM intensities (generally between 0.055 and 0.813 A m^{-1}), low coercivities and extremely low blocking temperatures (Fig. 5 *d* and *e*). NRM directions, directions of secondary components, and directions of components trending toward the origin in demagnetization are generally dispersed (Fig. 7). In AF demagnetization past 10 mT (100 Oe) or so, the behavior of the remanence is extremely erratic, with spurious components often acquired. Eastward-declination, shallow-inclination components are rare. In thermal demagnetization, over 95% of the remanence is removed by 300–330°C.

In general, only very irregular behavior occurs at higher temperatures. This character is best exemplified by samples from sites 31, 200, 202, and 204. It is not, however, restricted to a particular rock type, a general area, or a sampling site and this irregularity has an important bearing on the nature of the difference in magnetic properties of apparently similar lithologies such as the Fe-rich gabbros. For example, 14 of the 17 samples from site 202 have high NRM intensities (0.045–0.430 A m^{-1}) and display normal group 2 behavior in demagnetization. Samples 202G, 202O, and 202P, however, have NRM intensities of 0.0027, 0.0054, and 0.0044 A m^{-1}, respectively, show group 1 type behavior in demagnetization (Fig. 5*f*), and possess a high-coercivity, high-blocking temperature component of westward declination and shallow inclination. These three samples, as well as some samples from other sites whose remanence belongs dominantly to the group 2 type, are therefore group 1 samples. Another relatively consistent component is isolated in thermal demagnetization between 350 and 580°C (Fig. 6*a*). When resolvable, this component is invariably of westward declination and shallow inclination.

Group 3 samples

Group 3 samples are characterized by variable NRM intensities (0.430 A m^{-1} in serpentinites to 0.0050 A m^{-1} in Fe-gabbros), moderate to low coercivities, and generally uniformly distributed blocking temperatures (Fig. 6 *b–d*). Most importantly, these samples do not display a sharp decrease in magnetization intensity below 330°C (Fig. 6). This behavior is typical of samples from sites 32, 33, 34, 39, and 201. Although more dispersed than the characteristic remanence of group 1 samples, group 3 magnetization components trending toward the origin in demagnetization show a tendency to cluster in the northwest quadrant (Fig. 8*c*). Again, secondary components present are widely scattered (Fig. 8*b*).

In some group 3 samples, thermal demagnetization

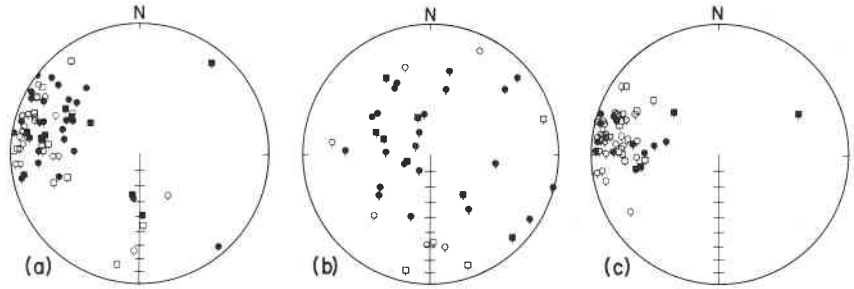


FIG. 4. Paleomagnetic data (*in situ*, equal-area net) from group 1 Ghost Range samples. Open (closed) symbols refer to projections on the upper (lower) hemisphere. (*a*) NRM direction; (*b*) any secondary direction; (*c*) characteristic direction. Squares (circles) refer to directions resolved in thermal (AF) demagnetization. Symbols with tick marks refer to directions from sites 36 and 37.

results in only a linear decay of a single component to the origin, whereas AF demagnetization resolves two components (Fig. 6 *e* and *f*). The blocking-temperature spectrum of the west-declination characteristic component overlaps significantly with that of the lower blocking-temperature (secondary) component, whereas the coercivity spectra of the two components do not overlap. The low-coercivity component is probably a chemical remanent magnetization (CRM), residing in secondary magnetite. If it were a partial thermoremanent magnetization (TRM), we would predict it to be resolved successfully in thermal as well as AF demagnetizations.

Magnetic mineralogy and rock magnetism

Response of the Ghost Range samples to AF and thermal demagnetizations supports the classification into three basic remanence types just discussed (Fig. 9). This classification is evident from the rock magnetization data in Table 1. Thermomagnetic studies (Fig. 10) and reflected-light examinations (Fig. 11) reveal that group 1 and group 3 remanences are carried by low-titanium magnetite (Fig. 10), whereas most of the remanence in group 2 samples resides in pyrrhotite. Curie temperatures of pyrrhotite in group 2 samples range between 290 and 320°C. These values, together with the smooth, simple shapes of the thermomagnetic curves at temperatures less than 350°C (Fig. 10), imply that the pyrrhotite is of Fe_7S_8 composition (Schwarz and Vaughan 1972; Schwarz 1975; Bennett and Graham 1980). In several instances (e.g., sample 200E, Fig. 10) the presence of pyrrhotite is not indicated by the cooling curves. The apparent loss of pyrrhotite is most likely due to the breakdown of initially ferrimagnetic Fe_7S_8 in heating. Textural evidence (e.g., euhedral to subhedral pyrrhotite inclusions in pyroxene and plagioclase, Fig. 11*a*) suggests that pyrrhotite is largely magmatic in origin, although in some cases large grains are spatially distributed along fractures and may be deuteric in origin.

Thermomagnetic analyses and reflected-light observations often reveal the presence of both Fe_7S_8 and low-titanium magnetite (Fig. 10). The equilibrium coexistence of pyrrhotite and magnetite is possible at temperatures in excess of 580°C under appropriate oxygen and sulfur fugacity conditions (e.g., Barton and Skinner 1967). It is concluded, therefore, that the consistent magnetization components present in group 2 samples above 330°C on thermal demagnetization reside in low-titanium magnetite, probably of magmatic origin.

Group 1 Fe-rich gabbros contain both coarse (generally greater than 100 μm) subhedral to irregular magnetite grains intergrown with mafic phases (Fig. 11*b*) and significant amounts of finer euhedral magnetite particles included in plagioclase and pyroxenes (Fig. 11 *c* and *d*), suggesting that these magnetite particles probably crystallized simultaneously with their hosts. The coarse, originally titaniferous, magnetite grains have invariably undergone both high-temperature oxidation–exsolution (Buddington and Lindsley 1964) during late-stage cooling, deuteric alteration, and later low-temperature oxidation probably during burial metamorphism. In the second stage of alteration, magnetite–ilmenite intergrowths were destroyed to varying degrees (Fig. 11*e*). Magnetite inclusions in silicate grains, on the other hand, are generally unoxidized. The high-blocking-temperature, high-coercivity remanence characteristic of group 1 samples probably resides in these fine, unaltered particles and is thus in all probability a TRM.

The magnetic mineralogy of group 3 samples also consists dominantly of magnetite, but petrographic observations correlate with the demagnetization behavior: the preserved magnetite grains are of a wider grain-size range and result in a gradual distribution of blocking temperatures.

Sulfur analyses

Naldrett and Goodwin (1977) and Naldrett *et al.*

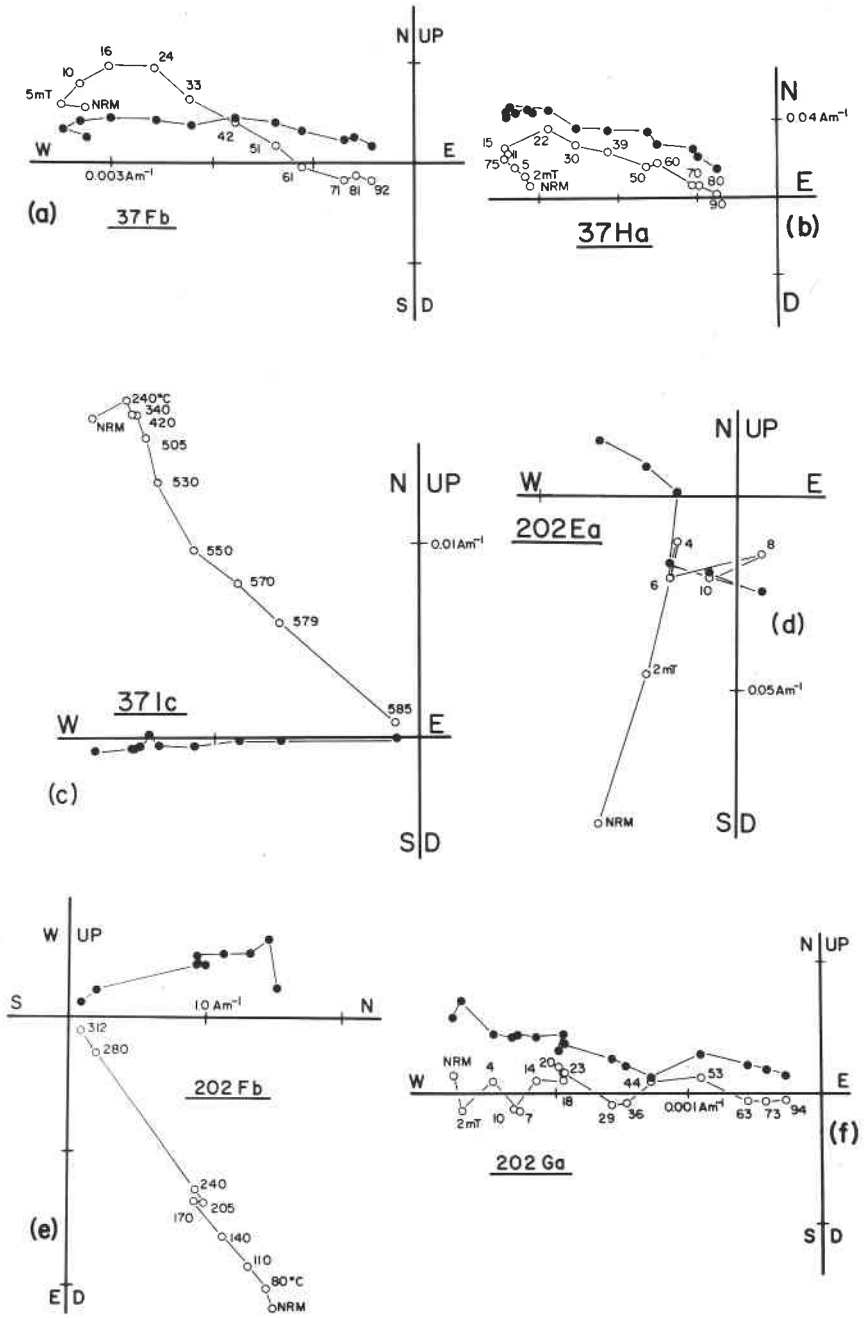


FIG. 5. Progressive orthogonal demagnetization diagrams (Zijderveld 1967) of the successive end point of the magnetization vector during demagnetization, plotted with respect to *in situ* coordinates. Solid (open) circles represent projections on the horizontal (east–west vertical) plane. Peak alternating fields (in mT) or temperatures (in C°) are given beside the vertical projections. Demagnetizations of (*a*, *b*, *c*) iron-rich gabbro (group 1) samples, site 37; (*d*, *e*) pyrrhotite-bearing iron-rich gabbro (group 2) samples, site 202; and (*f*) pyrrhotite-free iron-rich gabbro (group 1) sample, site 202.

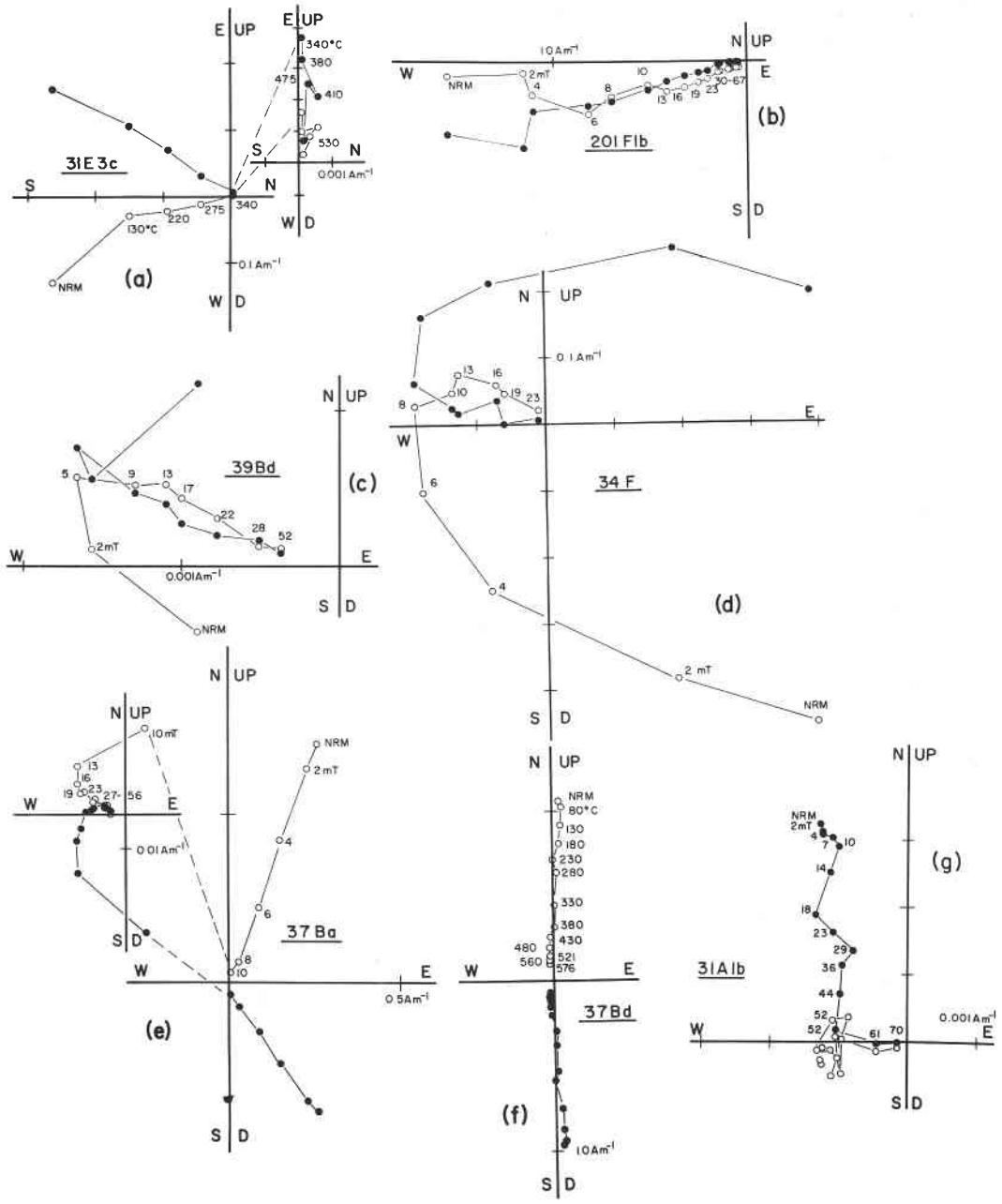


FIG. 6. Progressive orthogonal demagnetization diagrams (Zijderveld 1967). Conventions are as in Fig. 5. Demagnetizations of (*a*) pyrrhotite-bearing iron-rich gabbro (group 2) with higher blocking-temperature (group 3) remanence also, site 31; (*b*) magnesium-rich gabbro (group 3), site 201; (*c*) fine-grained gabbro (group 3), site 39; (*d*) serpentinite (group 3), site 34; (*e*, *f*) iron-rich gabbro (group 3) specimens, site 37; (*g*) iron-rich gabbro displaying a group 1 remanence composed of a lower coercivity Matachewan-dike-like component and a westerly component, site 31.

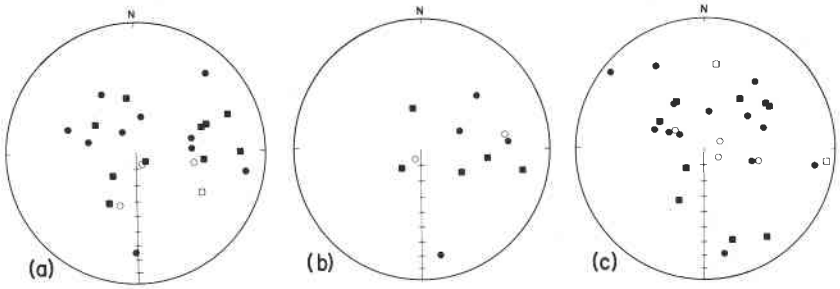


FIG. 7. Paleomagnetic data from group 2 Ghost Range samples. Conventions are as in Fig. 4. (*a*) NRM directions; (*b*) any secondary directions; and (*c*) low-blocking-temperature, low-coercivity directions residing predominantly in pyrrhotite.

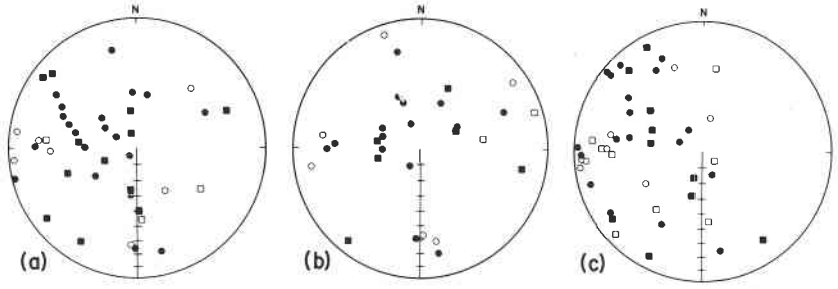


FIG. 8. Paleomagnetic data from group 3 Ghost Range samples. Conventions are as in Fig. 4. (*a*) NRM directions; (*b*) any secondary directions; (*c*) characteristic directions.

(1978) have reported on the sulfur content of Archean volcanic rocks in several greenstone belts of the Canadian Shield. Rocks of the central Abitibi Belt, belonging to the Stoughton–Roquemaure, Kinojevis, Blake River, and Timiskaming Groups, constituted the most extensive portion of their studies. For the Abitibi units, the arithmetic mean of the sulfur content of each lithologic subdivision was found to increase with mean FeO content. In addition, the arithmetic means of sulfur content are in agreement with experimental data on the sulfur content of sulfur-saturated basalts, suggesting that the Abitibi units were saturated with sulfur at the time of their extrusion.

For the Ghost Range lithologies, where sulfides are present, pyrrhotite is the dominant sulfide phase. We argue therefore that the relative variation in the sulfur concentration of the Ghost Range units should directly reflect the pyrrhotite abundance in these units. Accordingly, to test our explanation of the magnetic properties of the Ghost Range units, the sulfur content of representative samples of the Ghost Range lithologies was determined by X-ray fluorescence (XRF) methods. The samples analyzed were divided into three groups, according to their magnetic properties. The sulfur data are plotted in Fig. 12 and are consistent with our conclusions on the magnetic mineralogy inferred from paleomagnetic, petrographic, and rock magnetic observations. All group 1 samples have sulfur concentrations less than 300 ppm; all group 2 samples have sulfur concentrations greater than 400 ppm. In accord with our other observations, sulfur concentration of the Ghost Range lithologies is thus possibly an indirect, albeit crude, measure of their paleomagnetic reliability.

Discussion

The remanence data for the Ghost Range Complex are summarized in Table 2 and shown in Fig. 13*a*. The data in Table 2 are presented at both the specimen and the sample level. The most accurate representation of the paleomagnetic data from the Ghost Range is based upon giving unit weight to samples, rather than sites, as justified above. Generally, AF and thermal data were combined in the group directions, but in those instances where blocking-temperature spectra of magnetization components overlap, whereas as coercivity spectra do not (e.g., Fig. 6 *e* and *f*), only AF results are used to compute group directions. A westward-declination, shallow-inclination magnetization component can be detected in all the rock types in all sampled areas of the Ghost Range. In group 2 pyrrhotite-bearing samples this component could be present in fine-grained magnetite, but the presence of pyrrhotite obscures it. The high

TABLE 2. Summary of Ghost Range Complex paleomagnetic directions and poles; specimen and sample levels

Group	n/N	In situ D(°)	In situ I(°)	α_{95}(°)	k	VGP	Corrected D(°)	Corrected I(°)	Corrected VGP	Optimum demagnetization range (mT)	Optimum thermal demagnetization range (°C)
1	70/33	280	−4	6.0	8.9	1E, 5S (antipode)	279	−7	1E, 3S	20–90	500–585
	33 samples	280	2	11.8	5.5	13E, 7S (antipode)	287	2	8E, 12S		
2	23/17	75	66	37.9	1.6	19W, 42N	61	63	7W, 49N	NRM-10 (max.)	100–310
	17 samples	50	63	43.7	1.6	7E, 56N	40	58	24E, 60N		
3	42/24	274	13	18.1	2.5	12E, 8S (antipode)	276	12	10E, 9S	8–300	200–585
	24 samples	284	22	22.4	2.7	2E, 0N (antipode)	290	22	3W, 4S		

NOTES: n/N = the number of vectors (specimens) per number of independently oriented samples used to calculate mean direction; α_{95} = radius of circle of 95% confidence; k = estimate of precision parameter; VGP = virtual geomagnetic pole position.

FIG. 9. Normalized magnetization (*a*) versus peak alternating field during demagnetization and (*b*) versus temperature during demagnetization of representative Ghost Range samples.

electrical conductivity of Fe_7S_8 will screen the demagnetizing field and cause inhomogeneous demagnetization (Schwarz 1975) and will also cause the rock to acquire an ARM easily. At temperatures in excess of 300°C, Fe_7S_8 breaks down to Fe_9S_{10} and FeS_2 (depending upon atmospheric conditions). In turn, these phases may oxidize to magnetic phases capable of acquiring a moment, if the field is not perfectly nulled during progressive thermal treatments. Although even on the scale of an outcrop and within an individual rock unit (e.g., site 202) the magnetic properties and, therefore, distribution of pyrrhotite are irregular, the "background" remanence in samples containing only magnetite is well grouped.

If indeed the group 1 remanence resides in fine-grained, low-titanium, magnetic magnetite, then it was blocked well above the Curie temperature of Fe_7S_8, approximately 310°C (Schwarz and Vaughan 1972; Schwarz 1975). Furthermore, relations between the duration and maximum temperature of metamorphic and (or) burial reheating for the Ghost Range are such that

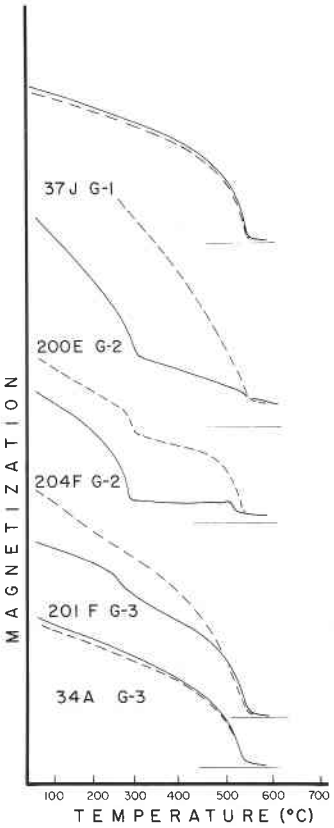


FIG. 10. Saturation magnetization versus temperature (thermomagnetic) curves of representative Ghost Range samples. Solid lines indicate heating curves; dashed lines indicate cooling curves.

the directions of the low-blocking-temperature, low-coercivity remanence in group 2 samples are not likely to reflect the geomagnetic field at the time of emplacement. The theoretical work of Pullaiah *et al.* (1975), pertaining to single-domain oxides, at least suggests that this remanence could have been blocked following a later thermal event and, because of the scattered directions, over a significantly longer period of time than the westward-declination characteristic component.

For several reasons we suggest that the west-declination magnetization component present in all lithologies of the Ghost Range is a TRM.

First, fine-grained magnetite inclusions, capable of carrying a TRM with the observed properties, have survived the oxidation that affected coarser grains.

Second, although the dominance of a single polarity of magnetization is not solid evidence for a TRM, it is in agreement with the findings of Dunlop (1980), who suggested polarity epochs of long duration during the Early Precambrian.

Third, the apparent polar wander (APW) path for the Precambrian of North America has been relatively well characterized back to the time of Matachewan dike intrusion (e.g., Irving 1979). The directions from the Matachewan dikes (2.6–2.7 Ga?) (Strangway 1964; Fahrig *et al.* 1965; Ernst and Halls 1980; Irving and Naldrett 1977), Abitibi dikes (2.1 Ga) (Fahrig *et al.* 1965; Schutts 1980; Ernst and Halls 1980; Irving and Naldrett 1977), and Hudsonian Orogeny and uplift (1.75–1.65 Ga) (Irving 1979) are shown calculated for Abitibi coordinates in Fig. 13*b*. Many, but by no means all, samples contain secondary components bearing strong resemblance to one of these tectonothermal events (e.g., the Matachewan-age component in sample 31A, Fig. 6*g*). The probable presence of magnetically dated superimposed components suggests that the higher coercivity and blocking-temperature westward-declination remanence predates these events.

Fourth, the westward-declination component, both *in situ* and corrected for slight structure (Fig. 13*a*), is in very good agreement with previously reported data from North America purported to represent the geomagnetic field for North America prior to Matachewan dike intrusion (Fig. 13*b*). These results are from the Kamiskotia Gabbro (KG) and Dundonald sill (DS), farther west of Munro Township in the Abitibi Belt (Irving and Naldrett 1977), the Shelly Lake Granite (SL) in the western Superior Province (A component, Dunlop and Buchan 1975), and scattered basalt and gabbro sites (BG) elsewhere in the central Abitibi Belt (Schutts 1980). *In situ* data from the Stillwater Complex, Montana (Bergh 1970) have previously been grouped into this data set (Irving and Naldrett 1977; Irving 1979). Because the Stillwater data should be corrected for Laramide tectonism, and recent Sm–Nd dates on the complex (De Paolo and Wasserburg 1979) have better constrained its age (2701 ± 8 Ma), we do not include Bergh's (1970) results in this pre-Matachewan Archean data set. The agreement between the Ghost Range data and previous results, however, is not itself evidence for a TRM, but rather an indication of the reliability of the data. We prefer to use the *in situ* direction from group 1 samples as representatives of the Ghost Range, as the slight northward tilt may be primary. At the present level of precision of the Late Archean data for North America, such an error is insignificant.

Fifth, we feel it is unreasonable to ascribe the high-coercivity, high-blocking-temperature remanence of the Ghost Range to significantly younger thermotectonic events within or near the Abitibi Belt whose associated magnetizations are very similar in direction to the pre-Matachewan one (e.g., Keweenawan igneous activity at approximately 1.15 Ga). There is little evidence for a magnetic remanence overprint or a petrolog-

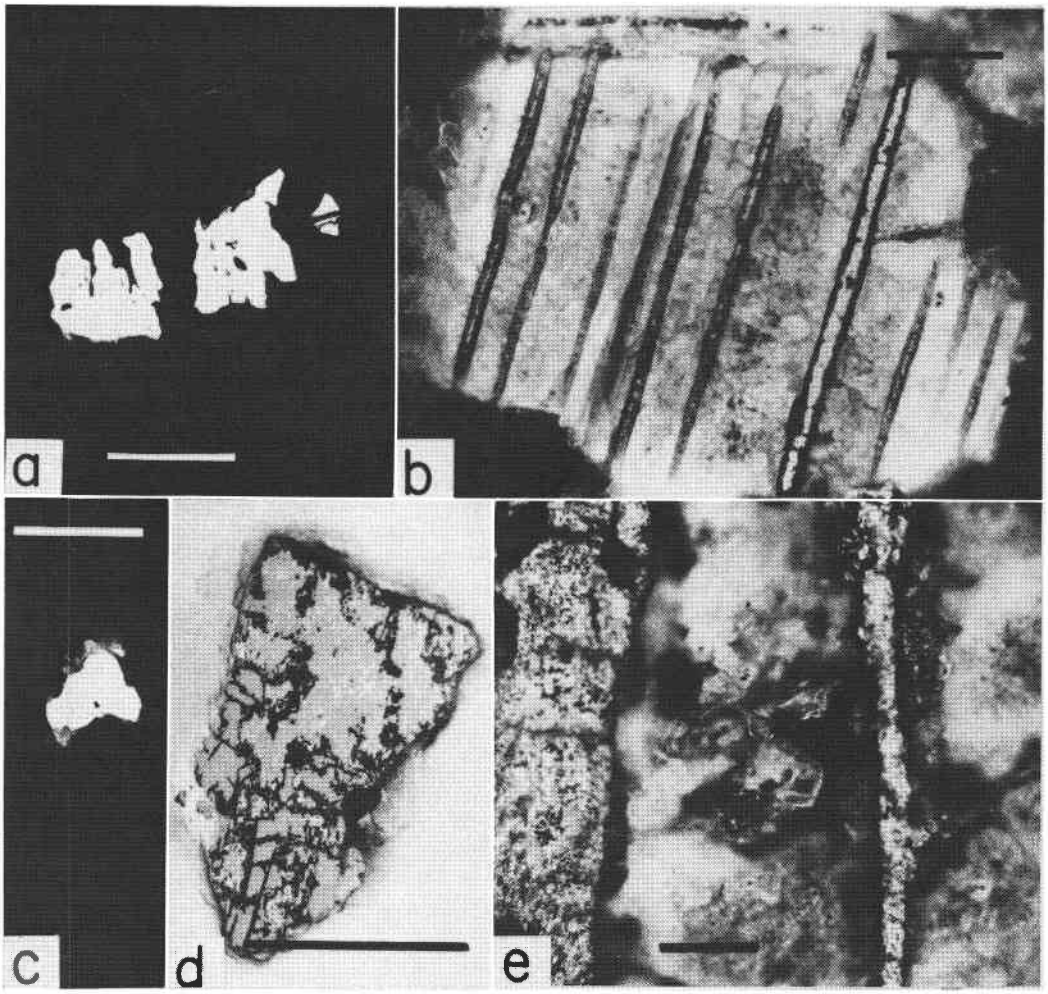


FIG. 11. Photomicrographs of representative magnetic oxide – sulfide parageneses in the Ghost Range lithologies. In each photomicrograph, the scale bar represents 20 μm. (*a*) Pyrrhotite inclusions in plagioclase, sample 202M, Fe-gabbro, pyrrhotite-bearing; (*b*) intensely altered magnetite–ilmenite intergrowth grain, sample 203E, Fe-gabbro, pyrrhotite-free; (*c*) fresh subhedral magnetite inclusion in plagioclase, sample 203E, Fe-gabbro; (*d*) slightly oxidized magnetite inclusion in clinopyroxene, sample 37B, Fe-gabbro, pyrrhotite-free; (*e*) enlargement of intensely altered magnetite–ilmenite intergrowth grain showing the textures exhibited by replacing hematite and rutile, sample 204F, Fe-gabbro, pyrrhotite-bearing.

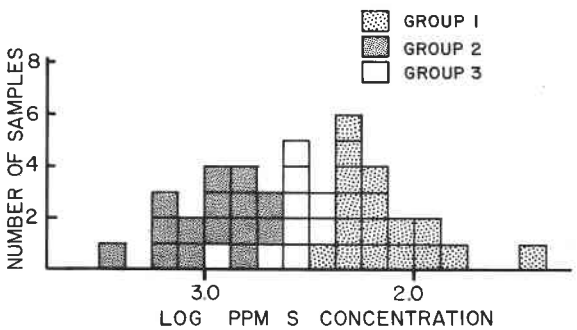


FIG. 12. Histogram of sulfur concentration data, Ghost Range magnetization groups.

ic–isotopic overprint of Keweenawan age in other units within or surrounding the Abitibi Belt (Schutts 1980; Geissman *et al.* in preparation; Tasillo-Hirt *et al.* 1982; Halls 1975; Hanes and York 1979). Exceptions include the results of Pesonen (1973) from Kinojevis basalts in Benoit and Melba Townships 30 km southwest of the Ghost Range. *In situ* data from strongly oxidized units on the north limb of a major east–west-trending synclinorium are statistically identical to those from group 1 and group 3 Ghost Range samples. Pesonen also found a superimposed secondary component of the same west-northwest declination in some samples from a Matachewan dike and adjacent, thermally reset Kinojevis flows exposed approximately 5 km north of the strongly

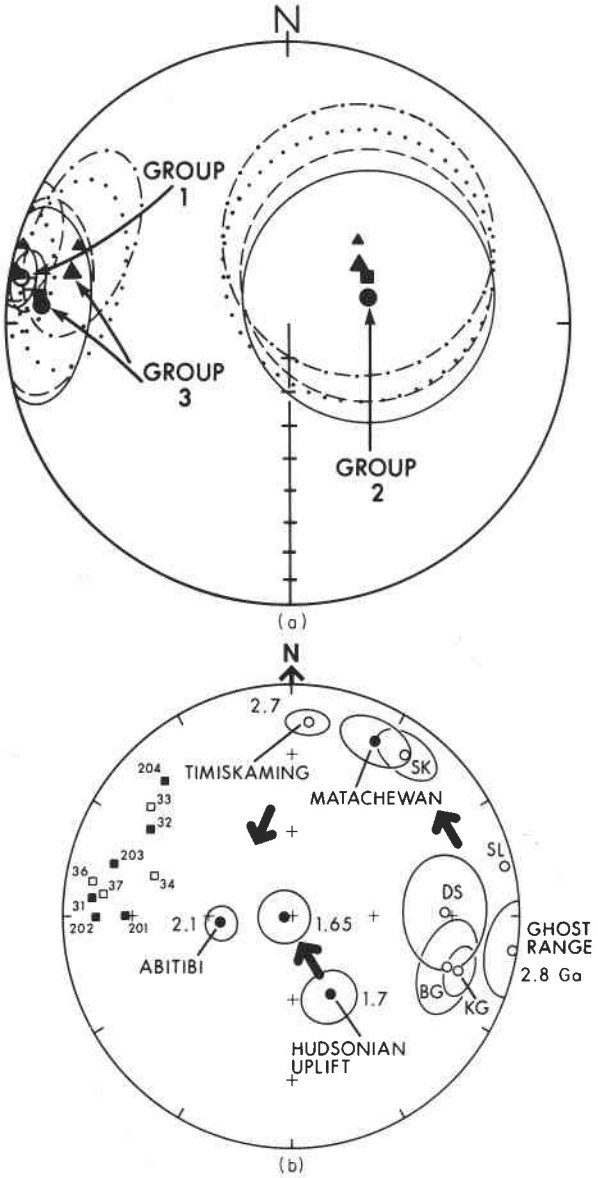


FIG. 13. (*a*) Summary of Ghost Range Complex paleomagnetic data. For specimen averages, *in situ* mean group directions are denoted by circles and solid projected cones of 95% confidence, structurally corrected mean group directions by squares and dashed projected cones of 95% confidence. For sample averages, *in situ* mean group directions are denoted by upright triangles and dotted projected cones of 95% confidence, structurally corrected mean group directions by inverted triangles and dot-dashed projected cones of 95% confidence. (*b*) Plot of paleomagnetic field directions expressed for central Abitibi coordinates (48°N, 80°W) (Abitibi coordinates) for Late Archean – Early Proterozoic time. KG = Kamiskotia Gabbro (Irving and Naldrett 1977); DS = Dundonald sill (Irving and Naldrett 1977); SL = Shelley Lake Granite, A component (Dunlop and Buchan 1976); BG = Abitibi basalts and gabbros (Schutts and Dunlop 1981); and Timiskaming = Timiskaming sediments (Tasillo-Hirt *et al.* 1982). Equal-area projections. Squares refer to *in situ* site means from the Ghost Range (Table 3).

oxidized flows. He argued that the west-northwest-declination component was of secondary origin, post-dating folding and probably acquired in Keweenawan time. Pesonen also recognized that this component is very localized, as he did not find it in other sampling sites nearby, and could not account for this phenomenon. Furthermore, such components were not found to be in statistical abundance by us in our examination of the Kinojevis and Blake River units surrounding Pesonen's (1973) study area (Geissman *et al.*, in preparation; Tasillo-Hirt *et al.* 1982).

Finally, the between-site distribution of westward-declination components (Table 3) is not unreasonable for a collection of geographically scattered sites in several lithologies. Jensen (1981) believed that the complex was emplaced relatively rapidly. As well, there are distinct time gaps in the succession of mafic magma pulses, as evidenced by the presence of cherty tuff horizons. The between-site dispersion of westward-declination components suggests that possibly some primary secular variation was recorded as a result of rapid emplacement, supporting our belief that the components were not acquired during a younger, uniform thermal overprint or over a long (greater than 10^{-4} to 10^{-5} years) duration of initial cooling.

The Ghost Range data provide additional evidence for the existence of a southeast- to east-declination, shallow-inclination magnetization direction for the Superior Province prior to Matachewan dike activity (Fig. 13). This direction converts into an equatorial paleopole off east-central Africa for the North American craton. As discussed earlier, the age of the complex is between 2700 and 2750 Ma. If the characteristic remanence of the Ghost Range is a TRM, as we propose, then the pre-Matachewan dike data in Fig. 13*b* indicate a rapid rate of APW (from 52°/10 Ma to 52°/200 Ma) of the Superior Province prior to Matachewan dike intrusion. The interpretation of a rapid migration is dependent on the Rb–Sr age given by Gates and Hurley (1973) for the Matachewan dikes. It would be useful to redate these dikes using the U–Pb zircon method or the samarium–neodynium method to better constrain their ages and this interpretation of the Archean APW path for North America.

Conclusion

A westward-declination, shallow-inclination magnetization component (D = 280°, I = 2°, k = 5.5, α_{95} = 11.8°, virtual geomagnetic pole (VGP) = 13°E, 7°S) residing in fine-grained magnetite has been isolated in

TABLE 3. Site magnetization data from the Ghost Range

Site	N	*In situ* D(°)	*In situ* I(°)	α_{95}(°)	k	Group 1 samples	Group 3 samples
31	7	274.7	11.3	40.4	3.2	2	5
32	6	300.5	26.4	25.1	8.1	6	—
33	7	307.0	−21.5	37.8	3.5	3	4
34	4	285.3	−36.3	68.6	2.8	—	4
36	5	279.4	−11.3	14.2	76.2	5	—
37	10	276.5	−16.2	7.6	41.4	8	2
201	6	270.8	28.0	34.6	4.7	—	6
202	3	269.5	19.0	32.5	15.4	3	—
203	7	285.9	18.4	21.3	8.9	7	—
204	1	316.4	19.7	—	—	—	1

NOTES: N = number of independently oriented samples used to calculate mean direction; α_{95} = radius of circle of 95% confidence; k = estimate of precision parameter.

the Ghost Range Complex of the central Abitibi Belt. This component is thought to be of TRM origin and, because the Ghost Range Complex is relatively undeformed, unmetamorphosed, and well dated, the results presented are a reliable contribution to understanding the pre-Mathachewan dike APW path for North America.

The results obtained from pyrrhotite-bearing samples confirm that the presence of pyrrhotite is a hindrance in paleomagnetic studies of the TRM of igneous rocks. We find it significant that a magnetite-dominated remanence, virtually identical to that from pyrrhotite-free sites, occurs in small, segregated volumes surrounded by pyrrhotite-bearing rock or where thermal demagnetization above 330°C is successful in isolating a magnetization component. In addition, this sporadic distribution of pyrrhotite in the Ghost Range units, confirmed positively by sulfur analyses, complicates the interpretation of magnetic anomalies over a terrane of potential economic interest. The magnetic properties of an individual lithology cannot be assumed to be constant.

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